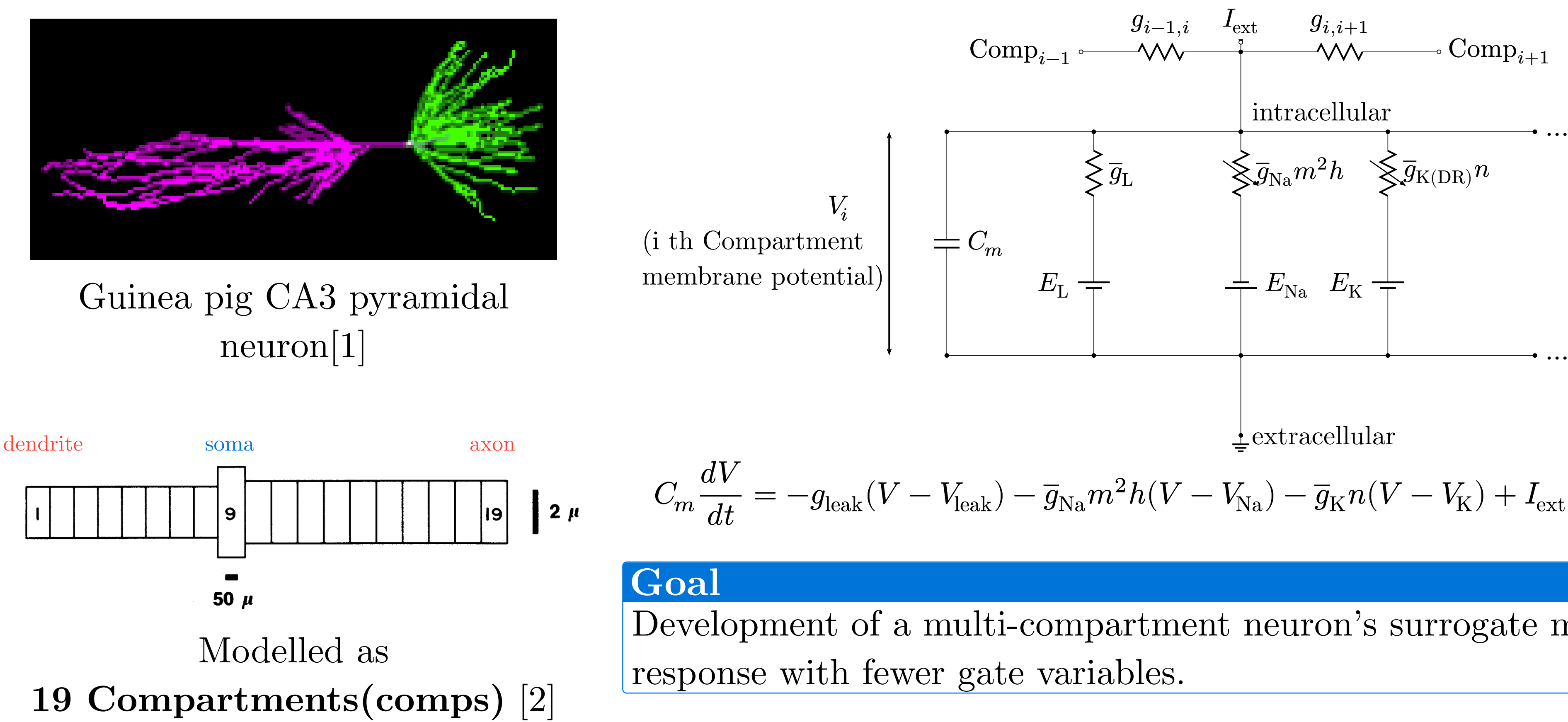


Introduction



Conductances have **gate variables**, ODEs with rate functions α, β :

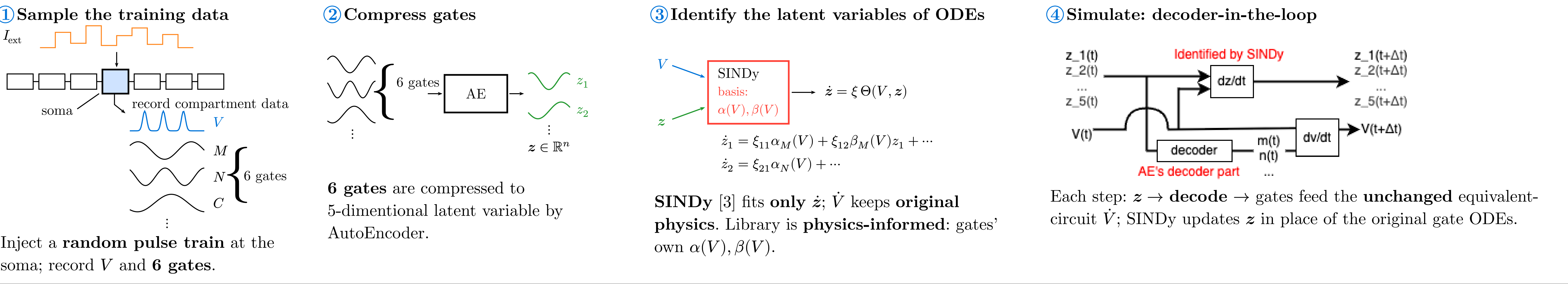
$$I_{\text{Na}} = \bar{g}_{\text{Na}} m^2 h (V - E_{\text{Na}})$$
$$\frac{dm}{dt} = \alpha_m(V)(1 - m) - \beta_m(V)m$$
$$\alpha_m(V) = 0.32(13.1 - u)/(\exp((13.1 - u)/4) - 1)$$
$$\beta_m(V) = 0.28(u - 40.1)/(\exp((u - 40.1)/5) - 1)$$

→ **11 states per compartment**
→ **209** for 19 comps;
At large scale network simulation, the gates become a **memory bottleneck**.

Goal

Development of a multi-compartment neuron’s surrogate model capable of reproducing the membrane potential response with fewer gate variables.

Methods



Results and Discussion

